

Module 1 — Read 2

Binomial test workflow and R

Module 1, Read 2: The binomial test in practice

Project: Project 1 — Testing a Claim · **Prerequisite:** Read 1 and Quiz A · **Next:** Quiz B, demo lab, exercise lab.

This reading is your **self-paced textbook** for conducting Project 1. Complete **Read 1** and Quiz A first. Use **Check yourself** boxes before **Quiz B**.

Learning outcomes

Three sections walk you from hypotheses through the full binomial test workflow in R.

After this reading you should be able to:

Section 1: Hypotheses and the binomial test

Introduction to the binomial test (ESP example)

- Translate a research question about a proportion into H_0 and H_a .
- Check the **four BRP conditions** for data from a fixed number of binary trials.
- Connect the ESP activity to the structure of **Project 1**.

Section 2: ESP worksheet workflow

ESP worksheet — binomial test in practice

- State hypotheses and verify the BRP conditions for the ESP class example.
- Identify n and p under H_0 and record the observed count x .
- Obtain a **p-value** in R using the same workflow as **Project 1**.

Section 3: Ten-step workflow and worked examples

Binomial test — steps and worked examples

- Carry out the full workflow: question, parameter, hypotheses, n and p under H_0 , BRP check, null distribution plot, observed x , shading by H_a , p-value, and conclusion in context.
- Handle **one-sided** and **two-sided** alternatives correctly when shading and interpreting the p-value.

 Do not use ESP as your Project 1 topic

The ESP examples below are **class models only**. Choose your own binomial process ($n = 25$) that is not a lecture example.

Section 1: Hypotheses and the binomial test

A motivating question: Do I have ESP?

Extra Sensory Perception (ESP) is the ability to accurately predict what will happen using supernatural methods. Do I have it? If so, I could make lots of money by betting on the stock market! To find out, we can take an online ESP test and then analyze the data using the **Binomial Test**.

The online ESP test

We will collect data from the **Advanced ESP (Zener Cards) test** at PsychicScience.org.

- The test uses **5 possible symbols** (circle, cross, waves, square, star).
 - On each trial, the computer randomly selects one symbol; your task is to guess it.
 - We will use **25 attempts** (25 cards).
 - Choose: **Clairvoyance, Open deck, Cards seen**, and **25 cards**.
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From question to hypotheses

! Notation: Parameter p and hypotheses

Let p be the probability of a correct prediction.

- If I **don't** have ESP, I'm just guessing among 5 options, so $p = 1/5 = 0.2$.
- If I **do** have ESP, I should get more than chance, so $p > 0.2$.

In conventional notation:

$$H_0 : p = 0.2 \quad \text{vs.} \quad H_a : p > 0.2$$

- **Null hypothesis** (H_0): baseline model with an **equality** (here, $p = 0.2$).
 - **Alternative hypothesis** (H_a): research question as an **inequality** (here, $p > 0.2$).
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Data collection and the four conditions

When we record the number of correct predictions out of 25 attempts, we treat this as a **Binomial Random Process**. We check the 4 conditions:

1. **Binary outcome:** Each prediction was either correct or not.
2. **Fixed number of trials:** We made a fixed number of attempts, $n = 25$.
3. **Fixed probability:** The probability of a correct guess on each trial doesn't change (open deck: each card is chosen at random from 5 options each time).
4. **Independent trials:** We don't use information from previous attempts to inform the next guess (and the computer randomizes each card independently).

So the number of correct predictions in 25 attempts can be modeled by a **binomial random variable** with $n = 25$ and $p = 0.2$ under the null hypothesis.

Sample project structure (Project 1)

The following is an example of how you might write up a **Binomial Test** in your one-page Project 1 summary. This example uses the ESP question and data from the online test.

Introduction

Extra Sensory Perception (ESP) is the ability to accurately predict what will happen using supernatural methods. Do I have it? If so, I could make lots of money by betting on the stock market! To find out, I took an online ESP test and then analyzed the data using the Binomial Test.

Let p be the probability that I have an accurate prediction. In the online ESP test, there were 5 possible predictions, so if I don't have ESP, then $p = 1/5$. If I do have ESP, then $p > 1/5$. These hypotheses in conventional notation are:

$$H_0 : p = 0.2 \quad \text{vs.} \quad H_a : p > 0.2$$

Data Collection

I recorded the number of correct predictions out of 25 attempts in the online ESP test at [Advanced ESP card guessing \(Zener Cards\) test — PsychicScience.org](#).

I believe this data collection satisfied the 4 conditions of a Binomial Random Process:

1. Each prediction was correct or not (**binary outcome**).
 2. I made a fixed number of attempts, $n = 25$ (**fixed number of trials**).
 3. My probability of getting a prediction correct shouldn't change between attempts (**fixed probability of success** in a single trial).
 4. I didn't use information from my previous attempts to inform my predictions (**independent trials**).
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Data Analysis

Descriptive statistics: In 25 attempts, I correctly predicted the shape on 4 cards (16%).

Inferential statistics: Let X be the number of correct predictions in 25 attempts. Assuming I'm just guessing, X is a Binomial random variable with $n = 25$ and $p = 0.2$. I computed the probability of each value of X and the probability of seeing my data (4) **or more** correct predictions using R. This **p-value** is 0.76.

Conclusion

There is a 76% chance of seeing my results of 4 correct predictions or more in another set of 25 attempts if I don't have ESP and am just guessing. Since this is a large probability, I've concluded I don't have ESP and will find other ways besides betting to make money.

Takeaway

! Definition: p-value

Under the **null model** (H_0), the **p-value** is the probability of observing data **as or more extreme than** what you observed (in the direction of H_a), **assuming H_0 is true**.

- A **small** p-value suggests the null model is not consistent with the data.
 - A **large** p-value (as in this example) suggests the data are consistent with just guessing.
- Your Project 1 report should follow this same structure: Introduction (question, p , hypotheses), Data Collection (how you collected data and why the 4 BRP conditions hold), Data Analysis (descriptive and inferential statistics, including p-value), and Conclusion.

! Seven steps of the binomial test

1. **Define the parameter of interest in your research question, p :** Identify and define p . Here p is the **theoretical** probability of a “success” in a single trial for the population or process (e.g., the probability of a correct guess on one ESP card). p is a parameter of the process, not a value observed in your sample.
2. **State the null and alternative hypotheses using appropriate symbols:**
 - **Alternative hypothesis (H_a):** The **direction** (inequality) of your best guess for the range of values of p — your research question in mathematical form (e.g., $p > 0.2$, $p < 0.2$, or $p \neq 0.2$).
 - **Null hypothesis (H_0):** Take the alternative and change the inequality to an **equality** (e.g., if $H_a : p > 0.2$, then $H_0 : p = 0.2$). This is the baseline or “no effect” assumption.
3. **Assuming the null hypothesis is true, generate the null distribution of x for sample size n :** Build the probability distribution of the number of successes, x , you would expect to see in n trials if H_0 were true. This null distribution is used to compute the p-value.
4. **Verify that the 4 conditions of a Binomial Random Process (BRP) are satisfied by the data collection:** Confirm that (a) each trial has a **binary** outcome (success or failure), (b) the number of trials n is **fixed**, (c) the probability of success p is **fixed** on every trial, and (d) trials are **independent**.
5. **Identify the value of x from the dataset:** Record the **observed** number of successes in your n trials (e.g., number of correct ESP guesses out of 25).
6. **Determine whether the data are consistent with the null hypothesis by computing the p-value:** The **p-value** is the probability of observing **new** data (in n trials) that is equal to or **more extreme** (in the direction of the alternative) than the data you actually observed, **assuming the null hypothesis is true**. A small p-value suggests the observed data are inconsistent with the null; a large p-value suggests they are consistent.
7. **Make a conclusion in context:** Interpret the p-value and state a conclusion about your research question (e.g., whether there is evidence for ESP or whether the results are consistent with chance).

R code: Colab notebook

You can practice binomial test code in the [Module 1 demo lab](#) and [R reference](#). The **ESP sample report** code (class model only) is in the course repo:

[Sample ESP notebook](#) · [Open in Colab](#) · [sample PDF](#) :: {callout-tip icon=false} ## Check yourself — Hypotheses and the binomial test

1. ESP test: 25 trials, 5 symbols, guessing under H_0 . Write H_0 and H_a for testing ESP.
2. In words, what does p represent in Project 1?

3. State the definition of **p-value** for a binomial test.
 4. Why does the ESP data collection satisfy the four BRP conditions?
- ...

Answers — Hypotheses and the binomial test

1. $H_0 : p = 0.2$ vs $H_a : p > 0.2$ where p = probability of a correct guess on one trial.
 2. The **theoretical** probability of success on **one trial** for the population/process — not the observed $\hat{p}$ from your sample.
 3. Probability of data **as or more extreme than observed** (in direction of H_a), **assuming H_0 is true**.
 4. Binary (correct/incorrect); fixed $n = 25$; fixed $p = 0.2$ under null (open deck); independent guesses.
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Section 2: ESP worksheet workflow

Follow the steps below to walk through the ESP class example, compute a p-value in R, and see how the pieces fit [Project 1](#). This follows the same structure as Section 1 and the [sample report \(PDF\)](#).

Step 1: State your null and alternative hypotheses (ESP)

You are testing whether you have ESP using the same setup as the instructor ESP example in Section 1.

! Notation: Parameter p

- Let p = the probability that **you** correctly predict the symbol on a single trial.
- If you don't have ESP, you're guessing among 5 symbols, so $p = 0.2$. If you do have ESP, $p > 0.2$.

Your turn. Write the null and alternative hypotheses and double-check that they match the hypotheses in Section 1:

$H_0: p = 0.2$

$H_a: p > 0.2$

Step 2: Verify the 4 conditions of a Binomial Random Process (in context)

Before we treat the number of correct guesses as a binomial random variable, we need to check that the 4 BRP conditions hold for this data collection.

! Checklist: Four BRP conditions (in context)

Your turn. In the context of the online ESP test (25 attempts, 5 symbols, open deck, cards seen), briefly think through why each condition is satisfied:

1. **Binary outcome:** Each trial has only two outcomes. Why? (What are they?)
2. **Fixed number of trials:** What is n ? How do we know it's fixed?
3. **Fixed probability of success:** Under the null hypothesis that you are guessing, what is the value of p on each trial? Why doesn't it change from trial to trial in the "open deck" setup?
4. **Independent trials:** Why doesn't one guess affect the next? Of if you believe guesses are not independent, what could you do to collect data in a way that your guess on one attempt doesn't affect your guess on another attempt?

Model answers — BRP checklist (ESP)

1. **Binary:** each guess is correct or incorrect.

2. **Fixed n :** you commit to 25 attempts before starting.
3. **Fixed p :** under H_0 , each guess has probability 0.2 (open deck re-randomizes each trial).
4. **Independent:** guesses should not use prior cards; if worried, take breaks or blind yourself to prior results.

Step 3: Identify n and p (under the null)

Your turn. For this activity we will all use 25 attempts. If the null hypothesis is true that **you** don't have ESP then

- $n = 25$
- $p = 0.2$

Step 4: Collect your data — take the ESP test

1. Go to the **Advanced ESP (Zener Cards) test**: <https://psychicscience.org/esp3>.
2. Watch the ad to access the test for “free”,
3. Use the default options: **Clairvoyance, Open deck, Cards seen, 25 cards**.
4. Complete all 25 guesses.
5. Record your **number of correct predictions** (this is your observed value of x).

Your x (number of correct predictions out of 25): *(record your own result after taking the test)*

Step 5: Find your p-value using the sample R notebook

1. Open the **sample ESP notebook** (Colab) or use **Module 1 demo lab** Section D for the generic pattern.
2. In the notebook, locate the observed number of correct predictions (the instructor example used $x = 4$).
3. **Change all the values of x from 4 to your value of x** (from Step 4).
4. **Run all the code** in the notebook (e.g., Runtime → Run all, or run each cell in order).
5. Find the **p-value** reported by the notebook and record it.

Your p-value: *(from the Colab notebook after you enter your x)*

Step 6: Post your results on Canvas

Go to the **Canvas discussion thread** for this activity.

Post the following:

- Your x (number of correct predictions out of 25).
 - Your **p-value**. We'll see who in class is **least consistent** with the hypothesis of no ESP by seeing who has the **smallest p-value** (or equivalently, the **largest x**). This will be the person who is most likely to have ESP! :::
{.callout-tip icon=false} ## Check yourself — ESP worksheet workflow
1. Under the null (no ESP), what are n and p for the 25-trial Zener test?
 2. What is your observed x after the test? (Use your own data.)
 3. Name the four BRP conditions you must defend in your Project 1 report.

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Answers — ESP worksheet workflow

1. $n = 25$; $p = 0.2$.
2. Answers vary; x = your number of correct guesses out of 25.
3. Binary outcome; fixed n ; fixed p on each trial; independent trials.

Section 3: Ten-step workflow and worked examples

! Ten steps of the binomial test (detailed workflow)

For each problem, follow these steps:

1. State the research question.
2. Define the parameter of interest, p , in words.
3. State the null and alternative hypotheses using appropriate symbols.
4. Identify n and the value of p under H_0 (for the null distribution).
5. Verify that the 4 conditions of a Binomial Random Process (BRP) are satisfied by the data collection.
6. Assuming the null hypothesis is true, create a plot of x vs $\text{dbinom}(x, n, p)$ (the null distribution) in R — same pattern as Project 1 (`geom_col + dbinom` heights).
7. Identify the observed value of x from the dataset.
8. On the plot, add `geom_vline` at the observed x and shade the bar(s) on the **same side(s) as the inequality in H_a** (coral bars in the examples below).
9. Compute the **p-value**: the probability of observing data as or more extreme than the observed x in the direction of H_a , assuming H_0 is true. (For a two-sided H_a , use both tails.)
10. Make a conclusion in context (reject H_0 or fail to reject H_0 , and what that means for the research question).

! Definition: p-value (binomial test)

The **p-value** is the probability of observing data **as or more extreme than** the observed x **in the direction of H_a , assuming H_0 is true**. For a two-sided H_a , sum probabilities in **both tails**.

! How to read the shaded null plots

Each figure below is the **null distribution** (x vs $P(X = x)$ under H_0).

1. **Vertical dashed line** — your **observed** count x from the data (the cut at the data you collected).
2. **Coral (red) bars** — the **p-value region**: bar heights you **add** for counts at least as extreme as yours **on the same side(s) as the inequality in H_a** :
 - $H_a : p > p_0 \rightarrow$ shade the **right** tail ($x \geq$ observed).
 - $H_a : p < p_0 \rightarrow$ shade the **left** tail ($x \leq$ observed).
 - $H_a : p \neq p_0 \rightarrow$ shade **both** tails (low counts and high counts that are as or more extreme).
3. **Steelblue bars** — counts that are **not** in the p-value region (still part of the null model, but not “as extreme” in the direction of H_a).

The **p-value** equals the sum of $P(X = x)$ over the shaded bars. Use the same R pattern in **Project 1** and the **demo lab**. Figures render when this page is built; the R code is hidden below each plot.

! Notation in worked examples

- **Parameter p** : theoretical probability of success on one trial (define in words for your question).
- **Observed data x** : number of successes in n trials.

- **Hypotheses:** H_0 uses an **equality**; H_a uses $<$, $>$, or $\neq$.

Example 1:

Research question: Is the proportion of left-handed students at our school higher than the national rate of 10%?

Parameter: p = probability that a randomly chosen student from our school is left-handed (i.e., the proportion of left-handers in the population we are sampling).

Hypotheses: $H_0 : p = 0.10$ vs $H_a : p > 0.10$

n and p under H_0 : $n = 25$ students, $p = 0.10$

BRP conditions (brief): Binary (left-handed or not), fixed $n = 25$, fixed p under null, independent students.

Null plot: $H_a : p > 0.10$ — shade where $x \geq 6$ (same side as $>$ in the alternative). Dashed line at observed $x = 6$.

Code anatomy — Null plot with p-value shading (Project 1 pattern)

Part	Role
<code>n 25</code>	Number of trials under H_0
<code>p 0.10</code>	Success probability under H_0
<code>x_obs 6</code>	Observed count from your data
<code>in_pvalue</code>	Logical vector: TRUE where bar is in p-value region ($x \geq$ observed (right tail))
<code>geom_col(aes(fill = in_pvalue))</code>	Bar heights from <code>dbinom</code> ; color marks p-value region
<code>geom_vline(xintercept = x_obs)</code>	Vertical line at observed x
Sum of coral bar heights	Approximates the p-value (check with <code>pbinom</code>)

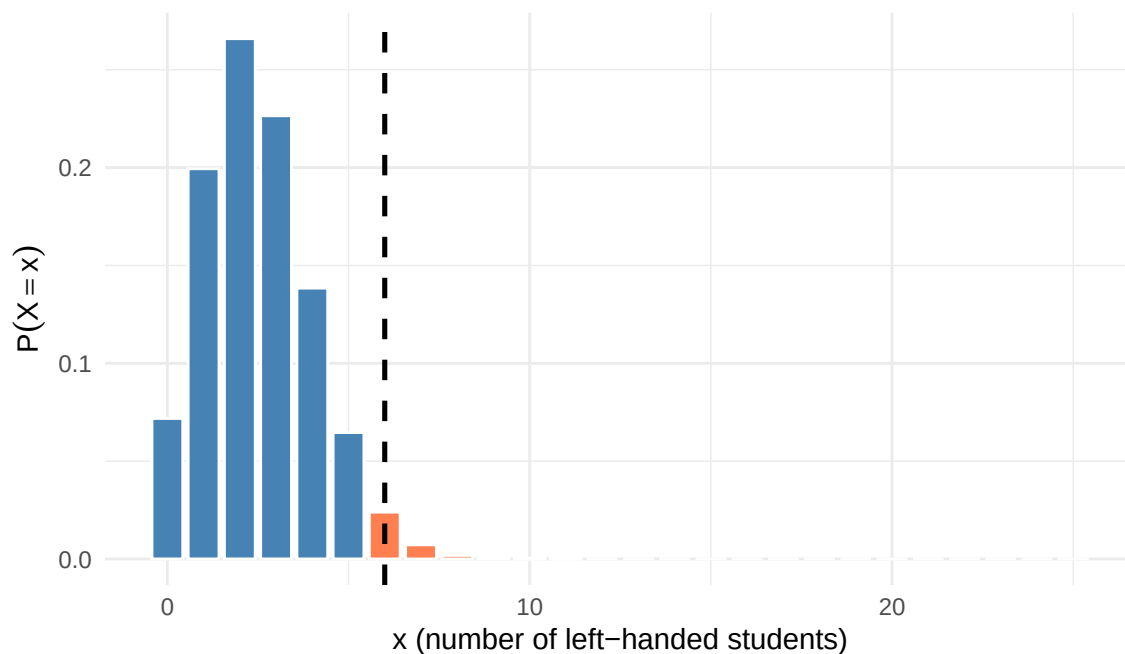


Figure 1: Example 1: left-handed students — p-value region shaded for $H_a : p > 0.10$, $x = 6$

Data: In a sample of 25 students, $x = 6$ were left-handed.

p-value: $P(X \geq 6 \mid H_0) = 1 - P(X \leq 5)$. For example, $1 - \text{pbinom}(5, 25, 0.10) \approx 0.03$. Small p-value.

Conclusion: Reject H_0 . The data are inconsistent with a 10% left-handed rate; we have evidence that the proportion of left-handed students at our school is higher than 10%.

Example 2

Research question: Is the proportion of defective items from the new supplier less than 15%?

Parameter: p = probability that a randomly selected item from the new supplier is defective (i.e., the proportion of defectives in the process).

Hypotheses: $H_0 : p = 0.15$ vs $H_a : p < 0.15$

n and p under H_0 : $n = 25$ items, $p = 0.15$

BRP conditions (brief): Binary (defective or not), fixed $n = 25$, fixed p under null, independent items.

Data: In a sample of 25 items, $x = 3$ were defective.

Null plot: $H_a : p < 0.15$ — shade where $x \leq 3$ (same side as $<$ in the alternative). Dashed line at observed $x = 3$.

Code anatomy — Null plot with p-value shading (Project 1 pattern)

Part	Role
n 25	Number of trials under H_0
p 0.15	Success probability under H_0
x_obs 3	Observed count from your data
in_pvalue	Logical vector: TRUE where bar is in p-value region ($x \leq$ observed (left tail))
geom_col(aes(fill = in_pvalue))	Bar heights from dbinom ; color marks p-value region
geom_vline(xintercept = x_obs)	Vertical line at observed x
Sum of coral bar heights	Approximates the p-value (check with pbinom)

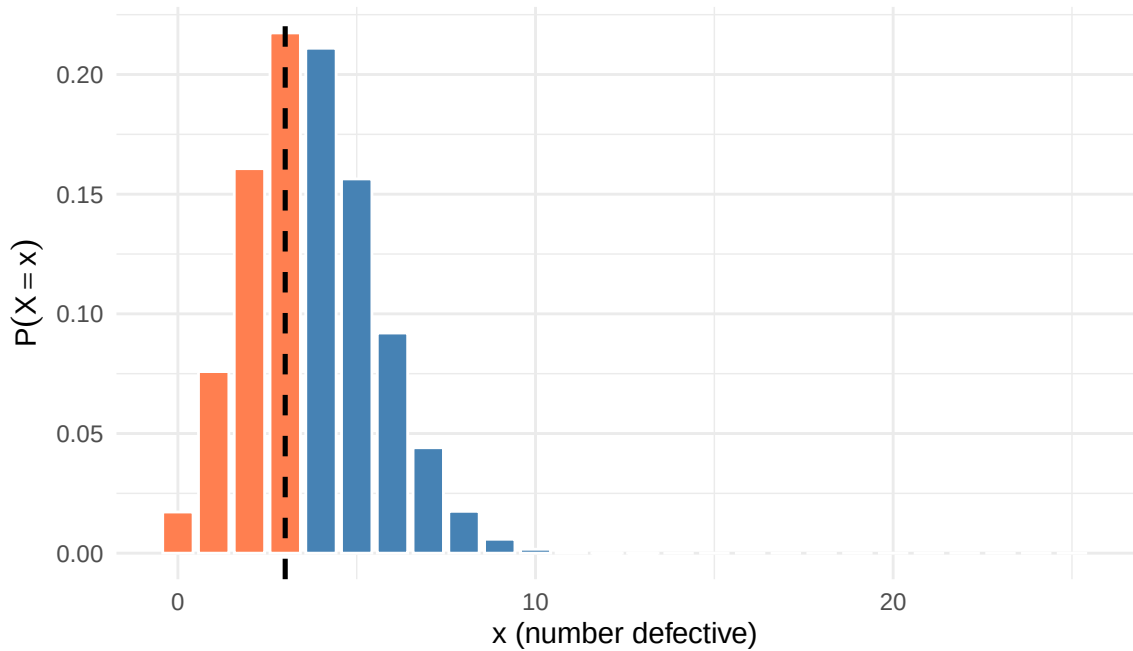


Figure 2: Example 2: defective items — p-value region shaded for $H_a: p < 0.15$, $x = 3$

p-value: $P(X \leq 3 \mid H_0) = \text{pbinom}(3, 25, 0.15) \approx 0.47$. Large p-value.

Conclusion: Fail to reject H_0 . The data are consistent with a 15% defective rate; we do not have evidence that the proportion of defectives from the new supplier is less than 15%.

Example 3

Research question: Are the two dice fair, in terms of the probability that the sum is 7 or 11? (With fair dice, that probability is 8/36: 6 ways to sum to 7 plus 2 ways to sum to 11, out of 36 equally likely outcomes.)

Parameter: p = probability that a single roll of two dice gives a sum of 7 or 11.

Hypotheses: $H_0 : p = 8/36$ vs $H_a : p \neq 8/36$

n and p under H_0 : $n = 25$ rolls of the two dice, $p = 8/36$ (under the null that the dice are fair).

BRP conditions (brief): Binary (sum is 7 or 11, or not), fixed $n = 25$, fixed p under null, independent rolls.

Data: In 25 rolls of the two dice, $x = 2$ times the sum was 7 or 11 (far from the expected $25 \times (8/36) \approx 5.6$ under H_0).

Null plot: $H_a : p \neq 8/36$ — shade **both** tails ($x \leq 2$ and $x \geq 9$ for this observed $x = 2$). Dashed line at observed $x = 2$.

Code anatomy — Null plot with p-value shading (Project 1 pattern)

Part	Role
n 25	Number of trials under H_0
p 8/36	Success probability under H_0
x_obs 2	Observed count from your data
in_pvalue	Logical vector: TRUE where bar is in p-value region (both tails for two-sided H_a)
geom_col(aes(fill = in_pvalue))	Bar heights from dbinom ; color marks p-value region
geom_vline(xintercept = x_obs)	Vertical line at observed x
Sum of coral bar heights	Approximates the p-value (check with pbinom)

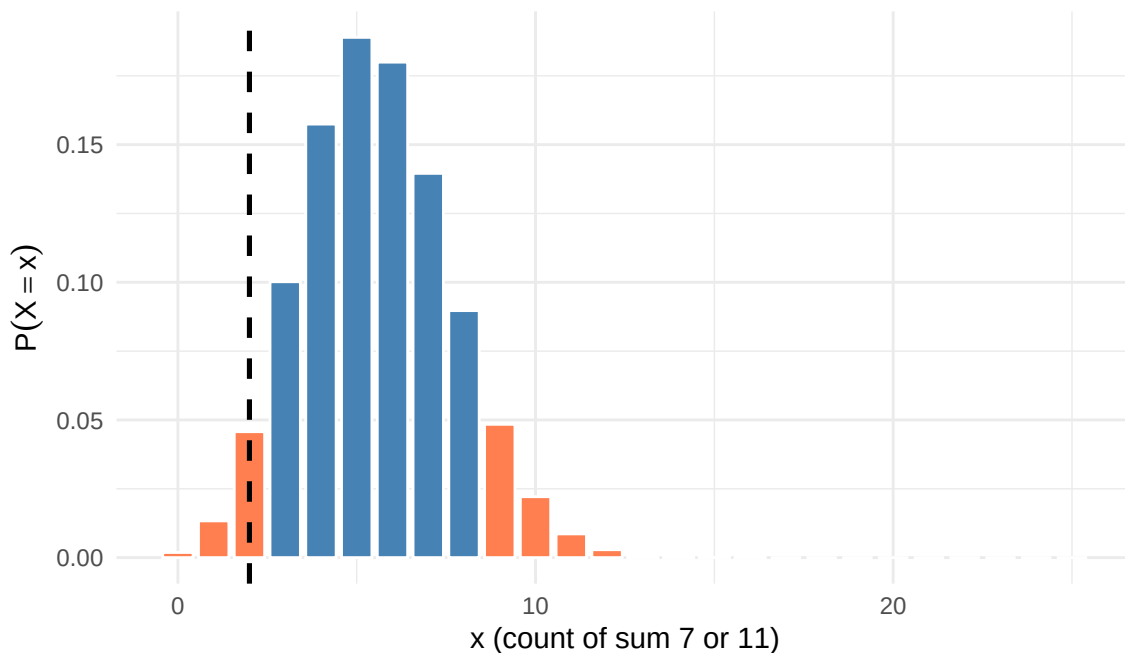


Figure 3: Example 3: dice sum 7 or 11 — two-sided p-value region shaded, $x = 2$

p-value: $P(X \leq 2) + P(X \geq 9)$ under H_0 . For example, `pbinom(2, 25, 8/36) + (1 - pbinom(8, 25, 8/36))` ≈ 0.04 . Small p-value.

Conclusion: Reject H_0 . The data are inconsistent with fair dice having probability $8/36$ for sum 7 or 11; we have evidence that $p \neq 8/36$.

Example 4

Research question: Do more than 20% of students at our school use the gym at least once per week?

Parameter: p = probability that a randomly chosen student uses the gym at least once per week (i.e., the proportion of such students in the population).

Hypotheses: $H_0 : p = 0.20$ vs $H_a : p > 0.20$

n and p under H_0 : $n = 25$ students, $p = 0.20$

BRP conditions (brief): Binary (uses gym weekly or not), fixed $n = 25$, fixed p under null, independent students.

Data: In a sample of 25 students, $x = 5$ use the gym at least once per week.

Null plot: $H_a : p > 0.20$ — shade where $x \geq 5$ (same side as $>$ in the alternative). Dashed line at observed $x = 5$.

Code anatomy — Null plot with p-value shading (Project 1 pattern)

Part	Role
<code>n 25</code>	Number of trials under H_0
<code>p 0.20</code>	Success probability under H_0
<code>x_obs 5</code>	Observed count from your data
<code>in_pvalue</code>	Logical vector: TRUE where bar is in p-value region ($x \geq$ observed (right tail))
<code>geom_col(aes(fill = in_pvalue))</code>	Bar heights from <code>dbinom</code> ; color marks p-value region
<code>geom_vline(xintercept = x_obs)</code>	Vertical line at observed x

Part	Role
Sum of coral bar heights	Approximates the p-value (check with <code>pbinom</code>)

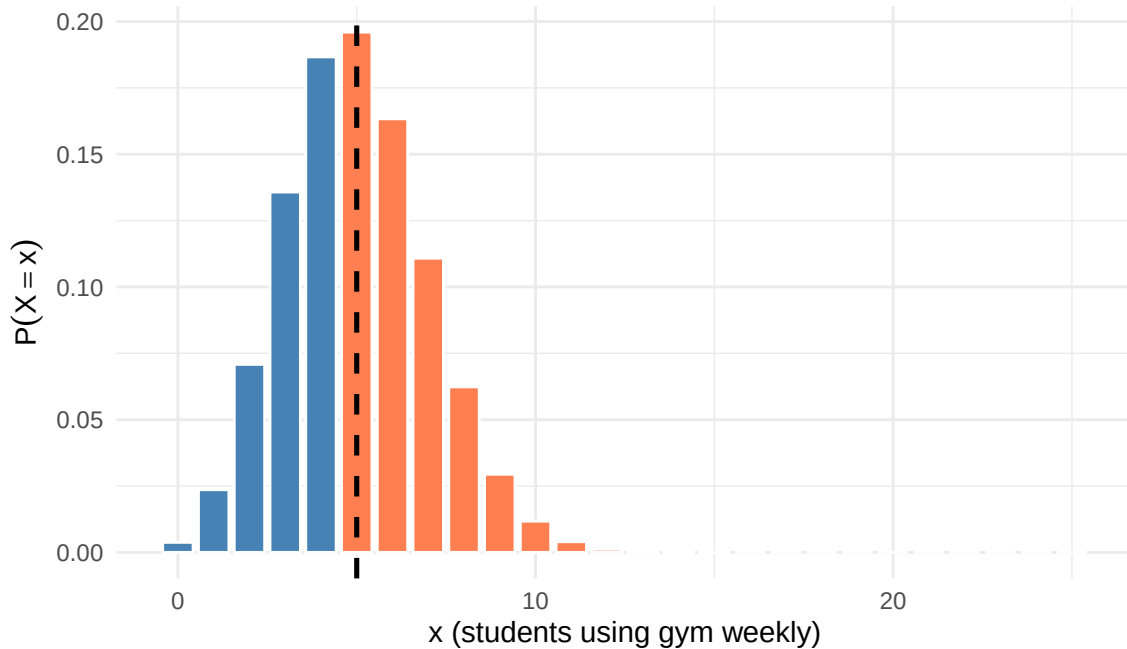


Figure 4: Example 4: gym use — p-value region shaded for $H_a: p > 0.20$, $x = 5$

p-value: $P(X \geq 5 \mid H_0) = 1 - P(X \leq 4)$. For example, $1 - \text{pbinom}(4, 25, 0.20) \approx 0.58$. Large p-value.

Conclusion: Fail to reject H_0 . The data are consistent with a 20% gym-use rate; we do not have evidence that more than 20% of students use the gym at least once per week.

Summary

Example	H_a direction	Observed x	p-value	Conclusion
1	$p > 0.10$	6	Small	Reject H_0
2	$p < 0.15$	3	Large	Fail to reject
3	$p \neq 8/36$ (two dice, sum 7 or 11)	2	Small	Reject H_0
4	$p > 0.20$	5	Large	Fail to reject

In every case, **steelblue** bars form the full null model, **coral** bars show the p-value region (matching the direction of H_a), and the **dashed line** marks observed x . The p-value is the sum of probabilities for the coral bars — compare that total to 1 when judging small vs large. ::: {callout-tip icon=false} ## Check yourself — Ten-step workflow and worked examples

1. Left-handed example: $H_0 : p = 0.10$ vs $H_a : p > 0.10$, observed $x = 6$. Which tail defines the p-value?
2. Defective items: $H_a : p < 0.15$, observed $x = 3$. Which tail defines the p-value?
3. Two-sided dice example: why shade **both** tails when $H_a : p \neq 8/36$?
4. What R function computes $P(X \leq k)$ for $X \sim \text{Binomial}(n, p)$?

...

Answers — Ten-step workflow and worked examples

1. Right tail: $P(X \geq 6 \mid H_0)$.
2. Left tail: $P(X \leq 3 \mid H_0)$.
3. “More extreme than” means far from the null in **either** direction.
4. `pbinom(k, n, p)` (with default `lower.tail = TRUE`).